

INSTITUTE OF ENGINEERING & MANAGEMENT

NEWTOWN, KOLKATA

Department of Electronics & Communication
Engineering

Module 2: Assignment :

Object detection models (YOLO, SSD, Faster R-CNN) on accuracy, speed, and suitability for real-time use in AVs

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1. Introduction

Object detection is an important computer vision task in autonomous vehicles. It enables a vehicle to identify objects such as pedestrians, cars, buses, and other road users. YOLO, SSD, and Faster R-CNN are commonly used object detection approaches with different characteristics in terms of accuracy and processing speed.

2. YOLO

YOLO (You Only Look Once) performs object detection in a single processing stage. It is known for its fast detection speed and is suitable for applications that require real-time processing.

Advantages:

- Fast detection speed
- Suitable for real-time applications
- Can detect multiple objects in a single image
- Suitable for video-based applications

Limitation:

- Detection accuracy can decrease for very small or closely positioned objects.

3. SSD

SSD (Single Shot Detector) is another single-stage object detection method. It uses multiple feature maps at different scales to detect objects of different sizes.

Advantages:

- Fast detection
- Suitable for real-time applications
- Can detect objects at different scales

Limitation:

- May have lower accuracy for small objects compared with some more complex detectors.

4. Faster R-CNN

Faster R-CNN is a two-stage object detection method. It first generates possible object regions and then classifies those regions.

Advantages:

- Generally provides strong detection accuracy
- Effective for detecting objects of different sizes
- Useful when detection accuracy is more important than speed

Limitation:

- More computationally demanding
- Slower than single-stage methods such as YOLO and SSD

5. Comparison

Feature	YOLO	SSD	Faster R-CNN
Detection approach	Single-stage	Single-stage	Two-stage
Speed	High	High	Lower
Accuracy	High	Good	High
Real-time suitability	Very suitable	Suitable	Less suitable
Computational requirement	Moderate	Moderate	High
AV application	Real-time perception	Real-time perception	Accuracy-focused detection

6. Suitability for Autonomous Vehicles

Autonomous vehicles need to detect road objects quickly because decisions such as braking, steering, and lane changes may depend on the detected objects. Therefore, detection speed is an important consideration along with accuracy. YOLO and SSD are suitable for real-time perception, while Faster R-CNN can be considered when higher computational resources are available and detection accuracy is given greater importance.

7. Conclusion

YOLO, SSD, and Faster R-CNN provide different approaches to object detection. YOLO and SSD focus on faster single-stage detection, while Faster R-CNN uses a two-stage approach. The appropriate model depends on the requirements of the autonomous vehicle system, available computational resources, and the required balance between detection speed and accuracy.